SUMMARY

The branch of science which deals the relation between energy, heat and work is called Thermodynamics. The main aim of the study of chemical thermodynamics is to learn (i) transformation of energy from one form into another form (ii) Utilization of various forms of energies.

System: A system is defined as any part of universe under consideration. There are three types of thermodynamic systems. They are (i) isolated system (ii) closed system and (iii) open system.

Surrounding: Everything in the universe that is not the part of the system is called surrounding.

Boundary: Anything which separates the system from its surrounding is called boundary.

Thermodynamic Properties: Any quantity that depends only on the state of system is called thermodynamic property of a system. There are two kinds of thermodynamic properties called (1) intensive - independent of the quantity of material and (2) extensive - directly proportional to the quantity of material. There are five basic thermodynamic properties. (U,H,S and G)

Thermodynamic Processes:

Adiabatic process in which no heat transfer takes place (q = 0)

Isothermal process in which temperature remains constant (dT =0).

Isobaric process in which pressure remains constant(dP =0).

Isochoric process in which volume remains constant(dV =0).

Cyclic process in which the system returns to its original state after completing a series of changes.

Internal energy (U): Internal energy of a system is equal to the energy possessed by all its constituents namely atoms, ions and molecules. The energy of a system of molecules is equal to the sum of its translational energy, vibrational energy, rotational energy, bond energy, electronic energy and energy due to molecular interactions.

Heat: Heat is regarded as the energy in transit across the boundary separating a system from its surrounding. Heat is a path function. The SI unit of heat is joule (J)

Work : Work is defined as the force (F) multiplied by the displacement -w=F.x, work is measured in Joules, i.e the SI unit of work is Joule. During expansion or compression of a gas the work done is calculated by the relation w= -PΔV.

The sign conventions for heat and work are as follows:

If heat is absorbed by the system : +q

11th Std Chemistry 186-228.indd 219 23-12-2021 15:33:59